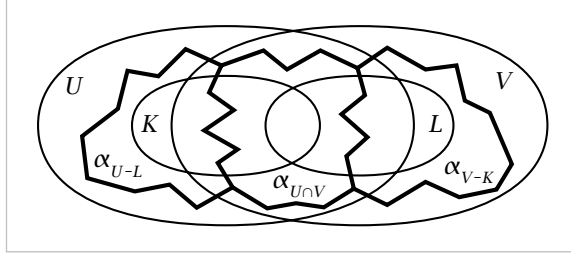


for $\varphi_A \in C^*(M, A)$ and $\varphi_B \in C^*(M, B)$. Then $\delta[\varphi]$ is represented by the cocycle $\delta\varphi_A = \delta\varphi_B \in C^*(M, A + B)$, where the equality $\delta\varphi_A = \delta\varphi_B$ comes from the fact that φ is a cocycle, so $\delta\varphi = \delta\varphi_A - \delta\varphi_B = 0$. Similarly, the boundary map ∂ in the homology Mayer-Vietoris sequence is obtained by representing an element of $H_i(M)$ by a cycle z that is a sum of chains $z_U \in C_i(U)$ and $z_V \in C_i(V)$, and then $\partial[z] = [\partial z_U]$.

Via barycentric subdivision, the class $\mu_{K \cup L}$ can be represented by a chain α that is a sum $\alpha_{U-L} + \alpha_{U \cap V} + \alpha_{V-K}$ of chains in $U - L$, $U \cap V$, and $V - K$, respectively, since these three open sets cover M . The chain $\alpha_{U \cap V}$ represents $\mu_{K \cap L}$ since the other two chains α_{U-L} and α_{V-K} lie in the complement of $K \cap L$, hence vanish in $H_n(M|K \cap L) \approx H_n(U \cap V|K \cap L)$. Similarly, $\alpha_{U-L} + \alpha_{U \cap V}$ represents μ_K .



In the square (*) let φ be a cocycle representing an element of $H^k(M|K \cup L)$. Under δ this maps to the cohomology class of $\delta\varphi_A$. Continuing on to $H_{n-k-1}(U \cap V)$ we obtain $\alpha_{U \cap V} \cap \delta\varphi_A$, which is in the same homology class as $\partial\alpha_{U \cap V} \cap \varphi_A$ since

$$\partial(\alpha_{U \cap V} \cap \varphi_A) = (-1)^k (\partial\alpha_{U \cap V} \cap \varphi_A - \alpha_{U \cap V} \cap \delta\varphi_A)$$

and $\alpha_{U \cap V} \cap \varphi_A$ is a chain in $U \cap V$.

Going around the square (*) the other way, φ maps first to $\alpha \cap \varphi$. To apply the Mayer-Vietoris boundary map ∂ to this, we first write $\alpha \cap \varphi$ as a sum of a chain in U and a chain in V :

$$\alpha \cap \varphi = (\alpha_{U-L} \cap \varphi) + (\alpha_{U \cap V} \cap \varphi + \alpha_{V-K} \cap \varphi)$$

Then we take the boundary of the first of these two chains, obtaining the homology class $[\partial(\alpha_{U-L} \cap \varphi)] \in H_{n-k-1}(U \cap V)$. To compare this with $[\partial\alpha_{U \cap V} \cap \varphi_A]$, we have

$$\begin{aligned} \partial(\alpha_{U-L} \cap \varphi) &= (-1)^k \partial\alpha_{U-L} \cap \varphi && \text{since } \delta\varphi = 0 \\ &= (-1)^k \partial\alpha_{U-L} \cap \varphi_A && \text{since } \partial\alpha_{U-L} \cap \varphi_B = 0, \varphi_B \text{ being} \\ &&& \text{zero on chains in } B = M - L \\ &= (-1)^{k+1} \partial\alpha_{U \cap V} \cap \varphi_A \end{aligned}$$

where this last equality comes from the fact that $\partial(\alpha_{U-L} + \alpha_{U \cap V}) \cap \varphi_A = 0$ since $\partial(\alpha_{U-L} + \alpha_{U \cap V})$ is a chain in $U - K$ by the earlier observation that $\alpha_{U-L} + \alpha_{U \cap V}$ represents μ_K , and φ_A vanishes on chains in $A = M - K$.

Thus the square (*) commutes up to a sign depending only on k . $\square$

Proof of Poincaré Duality: There are two inductive steps, finite and infinite:

(A) If M is the union of open sets U and V and if D_U , D_V , and $D_{U \cap V}$ are isomorphisms, then so is D_M . Via the five-lemma, this is immediate from the preceding lemma.

(B) If M is the union of a sequence of open sets $U_1 \subset U_2 \subset \cdots$ and each duality map $D_{U_i}: H_c^k(U_i) \rightarrow H_{n-k}(U_i)$ is an isomorphism, then so is D_M . To show this we notice first that by excision, $H_c^k(U_i)$ can be regarded as the limit of the groups $H^k(M|K)$ as K ranges over compact subsets of U_i . Then there are natural maps $H_c^k(U_i) \rightarrow H_c^k(U_{i+1})$ since the second of these groups is a limit over a larger collection of K 's. Thus we can form $\varinjlim H_c^k(U_i)$ which is obviously isomorphic to $H_c^k(M)$ since the compact sets in M are just the compact sets in all the U_i 's. By Proposition 3.33, $H_{n-k}(M) \approx \varinjlim H_{n-k}(U_i)$. The map D_M is thus the limit of the isomorphisms D_{U_i} , hence is an isomorphism.

Now after all these preliminaries we can prove the theorem in three easy steps:

- (1) The case $M = \mathbb{R}^n$ can be proved by regarding $\mathbb{R}^n$ as the interior of Δ^n , and then the map D_M can be identified with the map $H^k(\Delta^n, \partial\Delta^n) \rightarrow H_{n-k}(\Delta^n)$ given by cap product with a unit times the generator $[\Delta^n] \in H_n(\Delta^n, \partial\Delta^n)$ defined by the identity map of Δ^n , which is a relative cycle. The only nontrivial value of k is $k = n$, when the cap product map is an isomorphism since a generator of $H^n(\Delta^n, \partial\Delta^n) \approx \text{Hom}(H_n(\Delta^n, \partial\Delta^n), R)$ is represented by a cocycle φ taking the value 1 on Δ^n , so by the definition of cap product, $\Delta^n \frown \varphi$ is the last vertex of Δ^n , representing a generator of $H_0(\Delta^n)$.
- (2) More generally, D_M is an isomorphism for M an arbitrary open set in $\mathbb{R}^n$. To see this, first write M as a countable union of nonempty bounded convex open sets U_i , for example open balls, and let $V_i = \bigcup_{j < i} U_j$. Both V_i and $U_i \cap V_i$ are unions of $i-1$ bounded convex open sets, so by induction on the number of such sets in a cover we may assume that D_{V_i} and $D_{U_i \cap V_i}$ are isomorphisms. By (1), D_{U_i} is an isomorphism since U_i is homeomorphic to $\mathbb{R}^n$. Hence $D_{U_i \cup V_i}$ is an isomorphism by (A). Since M is the increasing union of the V_i 's and each D_{V_i} is an isomorphism, so is D_M by (B).
- (3) If M is a finite or countably infinite union of open sets U_i homeomorphic to $\mathbb{R}^n$, the theorem now follows by the argument in (2), with each appearance of the words 'bounded convex open set' replaced by 'open set in $\mathbb{R}^n$.' Thus the proof is finished for closed manifolds, as well as for all the noncompact manifolds one ever encounters in actual practice.

To handle a completely general noncompact manifold M we use a Zorn's Lemma argument. Consider the collection of open sets $U \subset M$ for which the duality maps D_U are isomorphisms. This collection is partially ordered by inclusion, and the union of every totally ordered subcollection is again in the collection by the argument in (B), which did not really use the hypothesis that the collection $\{U_i\}$ was indexed by the positive integers. Zorn's Lemma then implies that there exists a maximal open set U for which the theorem holds. If $U \neq M$, choose a point $x \in M - U$ and an open neighborhood V of x homeomorphic to $\mathbb{R}^n$. The theorem holds for V and $U \cap V$ by (1) and (2), and it holds for U by assumption, so by (A) it holds for $U \cup V$, contradicting the maximality of U . $\square$

|| **Corollary 3.37.** *A closed manifold of odd dimension has Euler characteristic zero.*

Proof: Let M be a closed n -manifold. If M is orientable, we have $\text{rank } H_i(M; \mathbb{Z}) = \text{rank } H^{n-i}(M; \mathbb{Z})$, which equals $\text{rank } H_{n-i}(M; \mathbb{Z})$ by the universal coefficient theorem. Thus if n is odd, all the terms of $\sum_i (-1)^i \text{rank } H_i(M; \mathbb{Z})$ cancel in pairs.

If M is not orientable we apply the same argument using $\mathbb{Z}_2$ coefficients, with $\text{rank } H_i(M; \mathbb{Z})$ replaced by $\dim H_i(M; \mathbb{Z}_2)$, the dimension as a vector space over $\mathbb{Z}_2$, to conclude that $\sum_i (-1)^i \dim H_i(M; \mathbb{Z}_2) = 0$. It remains to check that this alternating sum equals the Euler characteristic $\sum_i (-1)^i \text{rank } H_i(M; \mathbb{Z})$. We can do this by using the isomorphisms $H_i(M; \mathbb{Z}_2) \approx H^i(M; \mathbb{Z}_2)$ and applying the universal coefficient theorem for cohomology. Each $\mathbb{Z}$ summand of $H_i(M; \mathbb{Z})$ gives a $\mathbb{Z}_2$ summand of $H^i(M; \mathbb{Z}_2)$. Each $\mathbb{Z}_m$ summand of $H_i(M; \mathbb{Z})$ with m even gives $\mathbb{Z}_2$ summands of $H^i(M; \mathbb{Z}_2)$ and $H^{i+1}(M; \mathbb{Z}_2)$, whose contributions to $\sum_i (-1)^i \dim H_i(M; \mathbb{Z}_2)$ cancel. And $\mathbb{Z}_m$ summands of $H_i(M; \mathbb{Z})$ with m odd contribute nothing to $H^*(M; \mathbb{Z}_2)$. $\square$

Connection with Cup Product

Cup and cap product are related by the formula

$$(*) \quad \psi(\alpha \frown \varphi) = (\varphi \smile \psi)(\alpha)$$

for $\alpha \in C_{k+\ell}(X; R)$, $\varphi \in C^k(X; R)$, and $\psi \in C^\ell(X; R)$. This holds since for a singular $(k + \ell)$ -simplex $\sigma: \Delta^{k+\ell} \rightarrow X$ we have

$$\begin{aligned} \psi(\sigma \frown \varphi) &= \psi(\varphi(\sigma| [v_0, \dots, v_k]) \sigma| [v_k, \dots, v_{k+\ell}]) \\ &= \varphi(\sigma| [v_0, \dots, v_k]) \psi(\sigma| [v_k, \dots, v_{k+\ell}]) = (\varphi \smile \psi)(\sigma) \end{aligned}$$

The formula $(*)$ says that the map $\varphi \smile: C^\ell(X; R) \rightarrow C^{k+\ell}(X; R)$ is equal to the map $\text{Hom}_R(C_\ell(X; R), R) \rightarrow \text{Hom}_R(C_{k+\ell}(X; R), R)$ dual to $\frown \varphi$. Passing to homology and cohomology, we obtain the commutative diagram at the right. When the maps h are isomorphisms, for example when R is a field or when $R = \mathbb{Z}$ and the homology groups of X are free, then the map $\varphi \smile$ is the dual of $\frown \varphi$. Thus in these cases cup and cap product determine each other, at least if one assumes finite generation so that cohomology determines homology as well as vice versa. However, there are examples where cap and cup products are not equivalent when $R = \mathbb{Z}$ and there is torsion in homology.

By means of the formula $(*)$, Poincaré duality has nontrivial implications for the cup product structure of manifolds. For a closed R -orientable n -manifold M , consider the cup product pairing

$$H^k(M; R) \times H^{n-k}(M; R) \longrightarrow R, \quad (\varphi, \psi) \mapsto (\varphi \smile \psi)[M]$$